Fall detection with a single Doppler radar sensor and LSTM recurrent neural network

Takayuki Imamura
Graduate School of Engineering
Fukuoka University
Fukuoka 814-0180, Japan
td212003@cis.fukuoka-u.ac.jp

Vasily G. Moshnyaga
Dept. Electronics Eng.& CS
Fukuoka University
Fukuoka 814-0180, Japan
vasily@fukuoka-u.ac.jp

Koji Hashimoto
Dept. Electronics Eng.& CS
Fukuoka University
Fukuoka 814-0180, Japan
khashi@fukuoka-u.ac.jp

2022 IEEE 65th International Midwest Symposium on Circuits and Systems (MWSCAS) | 978-1-6654-0279-8/22/$31.00 ©2022 IEEE | DOI: 10.1109/MWSCAS54063.2022.9859430

***Abstract*—Falls are a serious health concern and a main cause of injuries among elders living independently at home. In this paper, we describe a new system for real-time automatic fall detection. In contrast to related formulations, the system employs a single continuous-wave micro-Doppler radar sensor to monitor a subject and Long Short-Term Memory based recurrent neural network (RNN) to identify falls from the time-frequency characteristics of the sensor's returns. It does not require extra hardware or big data set to classify abrupt and slow falls from non-fall actions with superior detection accuracy.**

***Keywords*— Doppler radar, sensor, machine learning, Long Short-Term Memory, RNN, fall detection**

I. INTRODUCTION

Falls are the leading cause of fatal and non-fatal injuries among older people. According to [1], just in the year 2020, 27.1% of all adults aged ≥ 65 years reported falling at least once (35 Million cases), 66.1% of all injures registered in this age group were caused by accidental falls, and 55.1% of all deaths from unintentional injury were fall-related. Elders living alone are especially vulnerable to falls as they might not be able to rise due to injury or declining muscular activity [2]. Therefore, devices capable of detecting falls and calling for support have become a necessity for elders living independently at home.

A large number of automatic fall detection technologies have been proposed over the last two decades. (See [3], [4] for surveys in the field). Wearable fall detectors are accurate and inexpensive but must be worn all the time, which is difficult for old and forgetful people. Therefore, context-aware systems with unobtrusive sensors embedded in the environment are considered more suitable for monitoring people at home. In this article, we focus on the accuracy improvement of Doppler-radar based context-aware fall detection systems.

Doppler-radar sensor is one of the most promising non-intrusive fall detection modalities: it transmits an electro-magnetic wave over a high-frequency range and measures the signal returns. In contrast to optical, ultrasonic and IR sensors, a Doppler-radar sensor outputs snapshots of the subject's velocities. It neither depends on the subject's occlusion, illumination, and distance, nor infringes human privacy. Also, unlike accelerometers, it neither requires direct contact with a subject nor control. However, robust identification of various types of falls based on Doppler radar is still a challenge.

The traditional approach to Doppler radar-based fall detection is to extract several hand-crafted discriminative features from the received signals and then apply them to fall classification algorithms. E.g. K-Nearest Neighbors [5], Kernel Matching Pursuit [6], Support Vector Machines [7], Bayesian Classifiers [8], Fuzzy integrals [9], and their combination [10]. Choosing the classification features is, however, a complicated task. It requires domain-specific knowledge of Doppler signatures and affects both the accuracy and scalability of the solution [11]. In addition, the features vary to a large extent with size, habits, and the health conditions of monitored individuals.

With advances in deep learning architectures and algorithms, recent efforts in Doppler radar-based fall detection have concentrated on statistical modelling of human falls by using deep learning techniques. In contrast to the traditional approach, these techniques rely on artificial neural networks that learn the most informative features and classification boundaries directly from input data without human intervention. Jokanovic et al. [12] used two stacked autoencoders and a logistic regression classifier to categorize falls. Tian et al [13] proposed to use two Doppler-radars and two neural networks to capture and process data in horizontal and vertical planes separately and then fuse the results to determine the spatiotemporal patterns of falls. These methods can distinguish an abrupt fall from no-fall motions, such as sitting, bending and walking. However, they either require expensive Range-Doppler radars [12] to capture both the frequency shift and the range or restrict themselves to multi-radar or multi-antenna installations [13-14]. Yoshino et al [15] presented a convolutional neural network (CNN) based on fall detection from signatures of a single continuous-wave Doppler radar that works in the conventional time-frequency domain. While their approach classifies abrupt falls quite well, it fails to recognize slow falls accurately.

In this work, we also use a single continuous-wave micro-Doppler radar sensor that operates in the frequency-time domain. However, contrary to [15], we employ a Long-Short-Term Memory (LSTM) based RNN to improve fall recognition. Compared to CNN, LSTM/RNN is more complex but better at learning long-term interdependences in data streams. In contrast to LSTM based fall detectors (e.g. [16-18]), which also rely on Doppler radars for people monitoring, our work does not require large data sets or multiple sensor installations. The article introduces a real-time LSTM based system that achieves robust fall detection with a single Doppler radar sensor.

978-1-6654-0279-8/22/$31.00 ©2022 IEEE
Authorized licensed use limited to: Universitas Brawijaya. Downloaded on September 01,2026 at 11:24:02 UTC from IEEE Xplore. Restrictions apply.

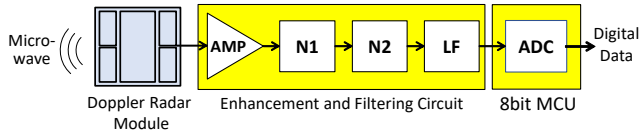


Figure 1. Block-diagram of Doppler radar sensor

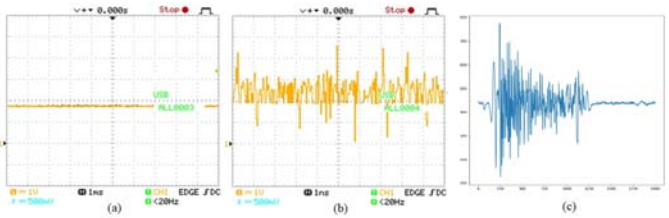


Figure 2. The signal waveforms at the output of the Doppler radar module

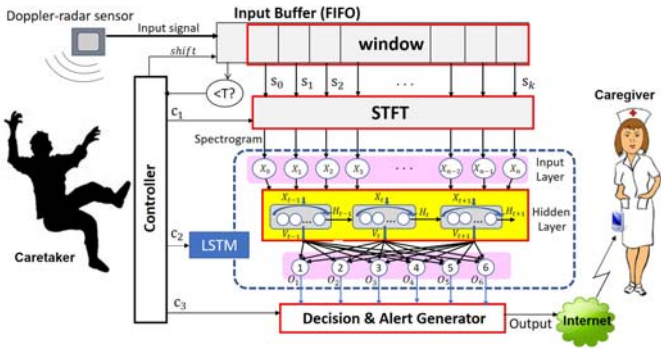


Figure 3. A block diagram of the proposed fall detection system

The main contribution of this work is twofold. First is a compact, inexpensive, and non-contact Doppler radar sensor, capable of capturing, filtering, enhancing, and delivering the frequency shift in a digital form with a high level of accuracy. The second is an LSTM-based fall detection system capable of recognizing different fall patterns from the outputs of a single Doppler radar sensor. Unlike related formulations, such as [16-18] that use LSTM and Doppler radar sensors to identify abrupt falls, the proposed system can distinguish fast and slow falls from non-fall actions, such as squatting, laying down, etc. The experiments show that the system works in real-time and has superior accuracy compared to related work.

II. The Proposed Fall Detection

The proposed fall detection system consists of a Doppler sensor, a small computer board and a server.

A. The Doppler radar sensor

We developed a compact and inexpensive Doppler sensor that outputs a signal of frequency proportional to the target's velocity and processes the signal; Namely, it magnifies, filters, and converts the analog signal into a digital representation.

Fig.1 shows the sensor's block diagram. The Doppler radar module contains a built-in patch antenna and an integrated IQ circuitry to compute a phase shift (Doppler frequency) between the transmitted (10.5GHz) signal and the reflected signal. The phase shift is then amplified by almost four orders of magnitude by the operational amplifier (AMP) and filtered

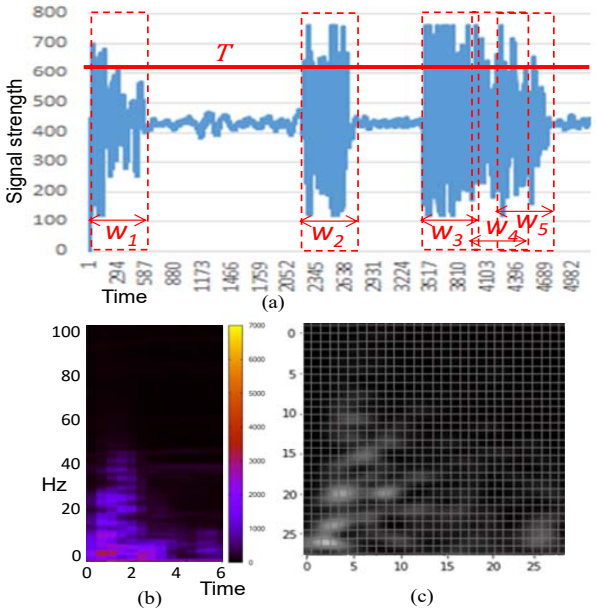


Figure 4 (a) An actual input signal with five activity intervals $w_1,\ldots, w_5$; (b) The spectrogram $X(w_1)$ of activities in the time interval w_1; (c) The normalized representation of spectrogram $X(w_1)$

by two band-stop/notch filters (N1 and N2) and the low-pass filter (LF) to reject block frequencies exceeding 60Hz as well as frequencies induced by the commercial power supply. Finally, the 8-bit microcontroller unit (MCU) converts the filtered analog signal (4.0Vp-p and 2.5V offset) into a 10 bits digital representation and sends it to the LSTM-based system for analysis. Fig.2 exemplifies the sensor performance by depicting the output signals of the Doppler module (Fig.2, a), the operational amplifier (Fig.2, b) and the ADC (Fig.2, c), respectively.

B. The LSTM-based fall-detection

Fig.3 shows the proposed LSTM based fall detection system. The system receives a 10-bit digital signal from the sensing unit and writes it in the FIFO buffer. If the signal amplitude overcomes a predefined threshold (T), the controller initiates the shift-right operation in the successive k-clock cycles to collect k+1-signals in the buffer. Else, it only compares the signal without shifting. After k clock cycles, the collected string of $k+1$ signals S_0 ... S_k is broadcast to the Short-Time Fourier Transform (STFT) to produce the time-frequency spectrogram, $X(w_i)$, for the string. The system cuts the DC-supply frequencies exceeding 60Hz, normalizes the values and outputs the spectrogram in 28x28 grayscale representation. Fig.4(a) illustrates the process on the example of 50s long signal stream captured by the Doppler Radar sensor. The stream reflects five intervals of the subject's activity (w_1-w_5). Fig. 4(b) shows the spectrogram, $X(w_1)$, obtained for the time interval w_1; Fig. 4(c) depicts the normalized representation of the spectrogram. The pixel brightness in the figures reflects velocity: the faster the motion, the brighter the pixel.

The normalized spectrogram representation is then sent to the LSTM/RNN classifier, which employs a seven-layer architecture (Fig.5, a) to identify a fall. (Note that dashed layers in the figure are used only in the training phase).

Authorized licensed use limited to: Universitas Brawijaya. Downloaded on September 01,2026 at 11:24:02 UTC from IEEE Xplore. Restrictions apply.

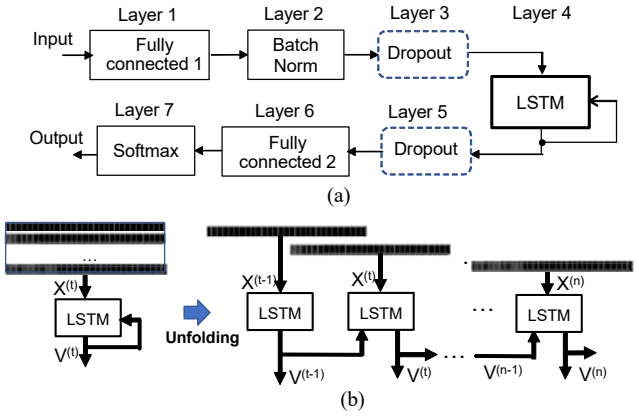


Figure 5. The LSTM/RNN architecture (a); temporal unfolding (b)

The classifier processes inputs in subsequences of fixed length (n=28) and cascades each cell into exactly n copies of itself by using temporal unfolding [19] with n/2 overlap between the subsequences. Fig.5 (b) illustrates the unfolding technique implemented in the classifier. Overall the architecture has 784 cells in the fully connected input layer, 128 cells in the LSTM (hidden) layer and six cells in the fully connected output layer.

The output cells of the LSTM correspond to six classes:

(1) abrupt fall, induced by balance loss when standing,

(2) abrupt motion-related fall, caused by a slip, a trip, or a stump,

(3) slow-fall or fading, caused by illness or intoxication, etc.

(4) squatting,

(5) lying down on the mat, and

(6) unspecified (i.e., all other no-fall actions, such as working, standing, jumping, etc.)

The probability scores of the classes are calculated by using the softmax and the cross-entropy error. The output with the maximum probability score determines the classification result for the current time window w_j.

After analyzing the time window w_i, the classification continues for w_{i+1}, and on. The system decides based on classification results obtained for an interval of 10s. If during this time it detects a fall, followed by a no-motion (i.e., the number of low-value inputs exceeds a given threshold), the alert generator (Fig.3) notifies the pre-specified caregiver with a message: "FALL! Help!" Otherwise, the system does not generate an alert. It only records the fall time and begins processing the next time-window (w_{i+1}) processing.

III. Implementation

The system is realized in hardware and software. Fig. 6 shows the photograph of the doppler sensor circuit. The circuit includes a 10MHz Doppler radar module (NJR4178J) from JRC, a low-cost programmable open-source platform (Arduino UNO) with an ATmega328P microcontroller on board, and several IC components in the enhancement and filtering circuit. The sensor communicates with a PC (HP Spectre x360, 2.8GHz i7 CPU, 16GB DRAM, Win10 OS) via a USB.

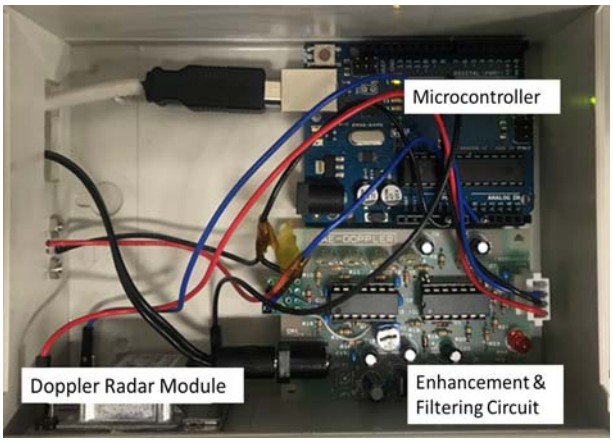


Figure 6. A photo of the Doppler sensor circuitry (view from the top)

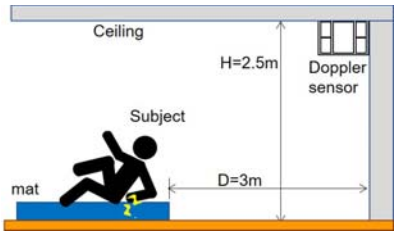


Figure 7. Experimental settings

To reduce the data transmission bandwidth between the sensor and the system, we use the ADC data sampling frequency of 100 Hz. The input buffer, STFT and LSTM/RNN and the decision and alert generator were implemented in software by using Python programming language and Pytorch v.19.0 library. Messages from the decision and alert generator to the caregiver's phone were conveyed via the LINE Notify free web-based service and LINE freeware application.

We trained the classifier on 1000 data samples labelled into batches by Pytorch, placed in Dataloder and applied as input for the RNN. To validate the classifier, we utilized 200 randomly selected samples. The validation results showed that the LSTM classifier had detection accuracy of 98% and 93% for abrupt and slow falls, respectively. After training, the LSTM/RNN classifier was uploaded into the system and tested.

IV. Experimental Evaluation and Results

We evaluated the system performance in laboratory settings with the sensor being set on the ceiling at the 2.5m height observing the room as shown in Fig. 7. Two subjects (male and female, both healthy adults) of 52–181cm height and 51-85 kg weight range took part in the experiments.

In the first experiment, each subject randomly performed the following sequence of actions: walking, falling (on the floor mat), standing, squatting, walking, slow falling (fading), lying on the flow for 5-8 seconds, standing, falling and no motion for 5-10 sec. All falls have been executed over the floor mat positioned at 3m (horizontal) distance (D) from the sensor's perpendicular point (see Fig.7). Each subject repeated the sequence five times. Thus, we tested 100 real-time actions in total.

Authorized licensed use limited to: Universitas Brawijaya. Downloaded on September 01,2026 at 11:24:02 UTC from IEEE Xplore. Restrictions apply.

TABLE I. CONFUSION MATRIX

True class	Output class						*Recall Score (%)*
	(1)	(2)	(3)	(4)	(5)	(6)	
(1)	100	0	0	0	0	0	100
(2)	0	97	3	0	0	0	97
(3)	5	0	93	0	2	0	93
(4)	0	0	0	99	0	1	99
(5)	0	0	3	0	97	0	97
(6)	0	2	0	4	0	94	94
Total	105	99	99	103	99	99	
Precision (%)	95.2	97.9	93.9	96.1	98.9	97.9	

Table I shows the results in terms of fall recognition *Precision* and *Recall* scores computed for the tested samples. As one can see, our system has an average precision score of 96.5% for abrupt falls (classes 1-2) and 93.9% for slow falls, while reaching a perfect (100%) recall score for abrupt falls with an average recall score of 96.6% for all classes.

The second experiment was dedicated to evaluating the impact of distance between the subject and the doppler sensor on the accuracy of the system. We experimented with the same setting (Fig. 7) and with the same subjects, who repeated the same sequence of actions for two distances (D=4m and D= 5m) from the mat to the Doppler sensor.

Table II shows the results in terms of recall score. Although the detection accuracy decreases with the distance from a subject to the sensor, the system provides a 91% recall score at a 5-meter distance and a 96.3% recall score on average at a 3 m distance.

TABLE II. DETECTION ACCURACY (%) VS. DISTANCE FROM SENSOR

Distance	Output class						Average Score
	(1)	(2)	(3)	(4)	(5)	(6)	
3 m	100	98	94	96	98	92	96.3
4 m	96	94	92	94	96	90	93.7
5 m	96	90	90	90	94	88	91.3

Table III compares this work with prior studies in terms of the classification model, the number of samples used for training the respective model, the number of patterns which can be recognized, and the average fall detection accuracy (Recall Score). In this table, AE abbreviates autoencoders, UWB is ultra-wide frequency modulated continuous band radar, FMCW is frequency modulated continuous wave radar. Note that [12], [13, [15] show simulation results on data sets, while the others provide results of real-time measurement. We observe that our system has better fall detection accuracy even though it was trained on a small data set and uses a single Doppler-radar sensor with one transmitter and one antenna.

TABLE III. COMPARISON WITH RELATED RESEARCH

Work	Classifier	Type & Number of sensors /antennas	Number of samples		Number of patterns		Average Fall Detection Acc. (%)
			falls	*non-fall*	*fall*	*non-fall*	
[12]	2 AE	1 range	408	117	4	3	88
[13]	2 CNN	2/2	550,541	541	18	40	92
[15]	CNN	FWCW 1/1	500	250	2	2	94
[16]	LSTM+CNN	UWB 3/3	600	60	1	1	88
[17]	LSTM	FMCW 2/2	711	3415	1	6	69
[18]	LSTM	FMCW 2/2	n/a	n/a	1	5	84[a]
Ours	LSTM	FMCW 1/1	600	400	3	3	95

[a] The bi-LSTM accuracy in radar only case

V. CONCLUSION

This article introduced a new automatic system to detect falls with a Doppler radar sensor. As the experiments revealed, the system identified falls accurately in real-time. At the same time, we acknowledge that the number of subjects, who participated in the experimental evaluation, was small and the testing was in the laboratory environment. Therefore, we now are extending the study to check system performance on more subjects, people of different body shapes, heights, and ages. Future studies will also focus on testing the system in residential settings (rooms, bathrooms, stairs, etc.)

REFERENCES

[1] America's Health Rankings analysis of CDC, Behavioral Risk Factor Surveillance System, United Health Foundation, AmericasHealthRankings.org, accessed 2022.

[2] Tinetti, et al. "Predictors and prognosis of inability to get up after falls among elderly persons". J. American Medical Association, 269 (1) Jan.1993, 65-70.

[3] M.Mubashir, et al. A survey on fall fetection: principles and approaches, Neurocomputing, vol.100, pp.144–152, Jan. 2013.

[4] T.Xu, et al. New advances and challenges of fall detection systems: A Survey. Applied Science, MDPI, 2018, 8, 418.

[5] L.Liu, et al. Fall detection using doppler radar and classifier fusion. IEEE-EMBS Int.Conf. Biomed.& Health Informatics, 2012

[6] Q.Wu, Y.D.Zhang, W.Tao, M.G.Amin. Fall Detection Based on Sequential Modeling of Radar Signal Time-Frequency Features, IEEE ICHI, 2013.

[7] Y.Kim and H.Ling, Human activity classification based on micro-Doppler signatures using a support vector machine, IEEE Trans. Geoscience & Remote Sensing (47) 5, pp.1328–1337, May 2009.

[8] Q.Wu, Y.Zhang, "Radar-based fall detection based on Doppler time–frequency signatures for assisted living". IET Radar Sonar Navig., vol. 9, issue 2, pp. 164–172, Feb. 2015

[9] M.Ren, J.Cai, Y.Zhu, J.Han. Radar signal feature extraction based on wavelet ridge and high order spectra analysis, IET Int. Radar Conf. 2009, pp.1-5.

[10] B.Erol, et al, Range-Doppler Radar Sensor Fusion for Fall Detection, Proc. IEEE Radar Conf. (RadarConf).

[11] M. Amin, et al. Radar Signal Processing for Elderly Fall Detection - The future for in-home monitoring, IEEE Signal Processing Magazine, March 2016, p.71-80.

[12] B.Jokanovic, et al. "Fall detection using deep learning in range-Doppler radars". IEEE Trans. Aerospace & Electronic Syst., vol.54, no.1, Feb.2018, pp.180-189.

[13] Y.Tian, et al. RF-based fall monitoring using convolutional neural networks, Proc.ACM Interact. Mob. Wearable Ubiquitos Technol. 2 (3), Art.137, Sep.2018.

[14] J. Hong, et al. Cooperative fall detection using doppler radar and array sensor. IEEE Int. Symp. Personal Indoor & Mobile Radio Comm. pp.3492–3496, 2013.

[15] H.Yoshino, et al. Fall detection on a single Doppler-radar sensor by using convolutional neural networks, IEEE SMC, pp.2875-2878, 2019.

[16] J.Maitre, et al.Fall Detection with UWB Radars and CNN-LSTM Architecture. IEEE J. Biomed.& Health Informatics, Oct. 2020, pp.1-11.

[17] Y.Sun, et al, Privacy preserving fall Detection with deep learning on mm wave radar signal. 2019 IEEE VCIP, pp.1-4.

[18] H. Li, et al. Bi-LSTM network for multimodal continuous human activity recognition and fall detection, IEEE Sensors J. vol.20, no.3 2019, pp.1191-1201

[19] Hochreiter, Sepp; Schmidhuber, Jürgen (1997). "Long Short-Term Memory". Neural Computation. 9 (8): 1735–1780.

Authorized licensed use limited to: Universitas Brawijaya. Downloaded on September 01,2026 at 11:24:02 UTC from IEEE Xplore. Restrictions apply.